


XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

# COM105 Engineering Physics (I)

Lecture 7: Work, Kinetic Energy,
Potential Energy, and Energy Conservation

**By:**
**Dr. Ng Zi Neng**
Email: zineng.ng@xmu.edu.my

# In this chapter, you'll learn…

- what it means for a force to do work on an object, and how to calculate the amount of work done.
- The definition of the kinetic energy (energy of motion) of an object, and how the total work done on an object changes the object's kinetic energy.
- How to use the relationship between total work and change in kinetic energy when the forces are not constant, the object follows a curved path, or both.
- how to solve problems involving power (the rate of doing work).

# Work and Energy

- In classical mechanics, we are concerned with the concepts of **work** and **energy**.
- Work-energy concepts are based on Newton's laws and do not involve any new physical principles.
- The work-energy concept can therefore be applied to the dynamics of a mechanical system **without** resorting to Newton's laws; useful in complex situations (e.g. variable forces) and applicable to a wide range of phenomena (e.g. electromagnetism and atomic physics)

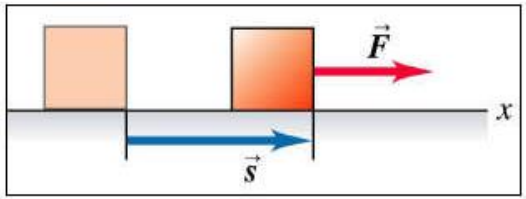
# Work and Energy

- A baseball pitcher does work with his throwing arm to give the ball a property called **kinetic energy**.
- In this chapter, the introduction of the new concepts of **work, energy, and the conservation of energy** will allow us to deal with problems in which Newton's laws alone aren't enough.



# Work Done by Constant Force

- Work done by force ≡ force × displacement (when force is parallel to displacement)

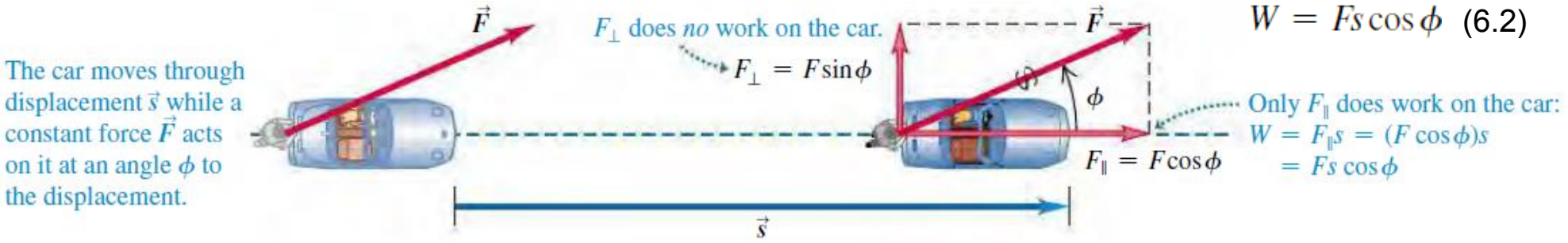



$$W = Fs \quad (6.1)$$

(constant force in direction of straight-line displacement)

SI Unit : N·m = joule (J)

- In general, a force on an object does work if the object undergoes a displacement.




$$W = Fs\cos\phi \quad (6.2)$$

# Work Done by Constant Force

Work done on a particle by constant force $\vec{F}$ during straight-line displacement $\vec{s}$

$$W = \vec{F} \cdot \vec{s} \qquad (6.3)$$

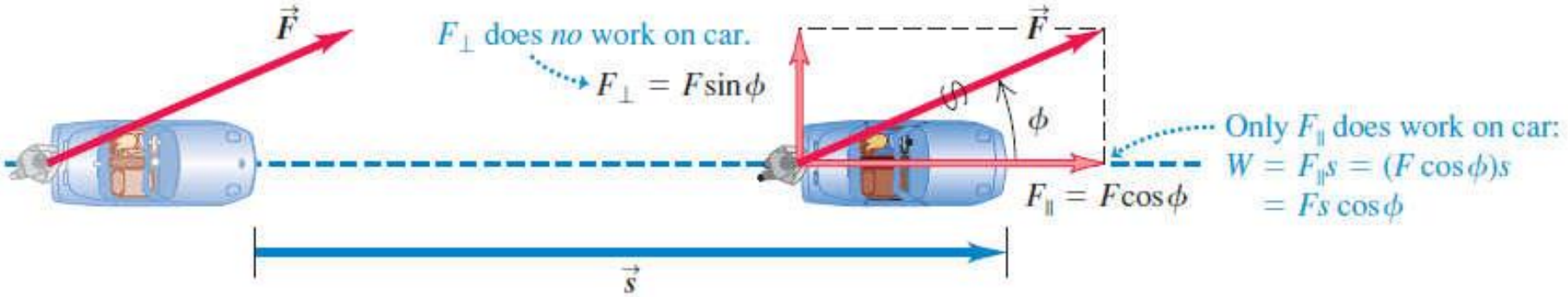
Scalar product (dot product) of vectors $\vec{F}$ and $\vec{s}$

- If Φ = 0, **F** and **s** are in the same direction, then cos Φ = 1 and we are back to Eq (6.1).
- A force does **no** work if the **object does not move**.
- Work done by a force is **zero** if the force is **perpendicular** (cos 90° = 0) to the displacement.
- Work can be **positive** or **negative**, e.g. when an object is lifted, the work done by the applied force is positive (lifting force is upward); work done by gravitational force however is negative.
- Work is a **scalar** quantity.

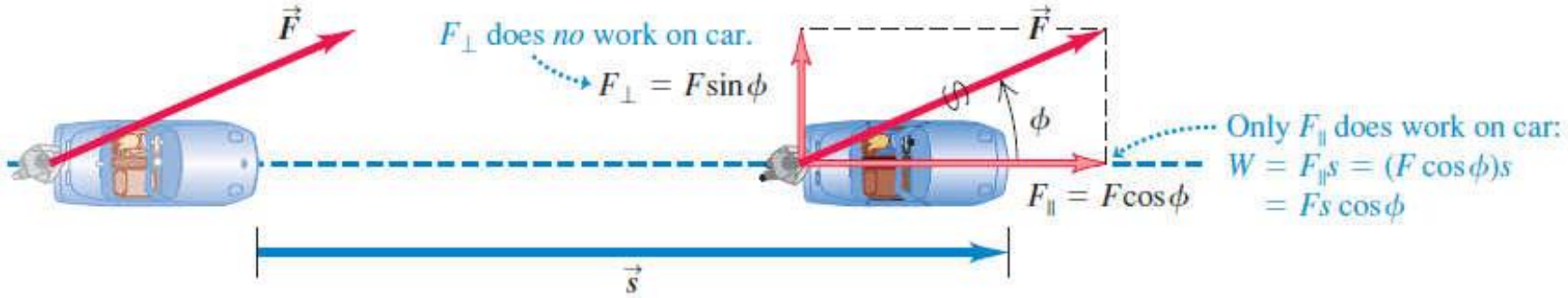
# Work Done by Constant Force

## Example 6.1 Work done by a constant force

(a) Steve exerts a steady force of magnitude 210 N (about 47 lb) on the stalled car in Fig. 6.3 as he pushes it a distance of 18 m. The car also has a flat tire, so to make the car track straight Steve must push at an angle of 30° to the direction of motion. How much work does Steve do? (b) In a helpful mood, Steve pushes a second stalled car with a steady force $\vec{F} = (160\ \text{N})\hat{\imath} - (40\ \text{N})\hat{\jmath}$. The displacement of the car is $\vec{s} = (14\ \text{m})\hat{\imath} + (11\ \text{m})\hat{\jmath}$. How much work does Steve do in this case?



# Work Done by Constant Force




**EXECUTE:** (a) From Eq. (6.2),

$$W = Fs\cos\phi = (210\text{ N})(18\text{ m})\cos 30^\circ = 3.3 \times 10^3\text{ J}$$

(b) The components of $\vec{F}$ are $F_x = 160$ N and $F_y = -40$ N, and the components of $\vec{s}$ are $x = 14$ m and $y = 11$ m. (There are no $z$-components for either vector.) Hence, using Eqs. (1.21) and (6.3), we have

$$W = \vec{F}\cdot\vec{s} = F_x x + F_y y$$

scalar product.

$$= (160\text{ N})(14\text{ m}) + (-40\text{ N})(11\text{ m})$$

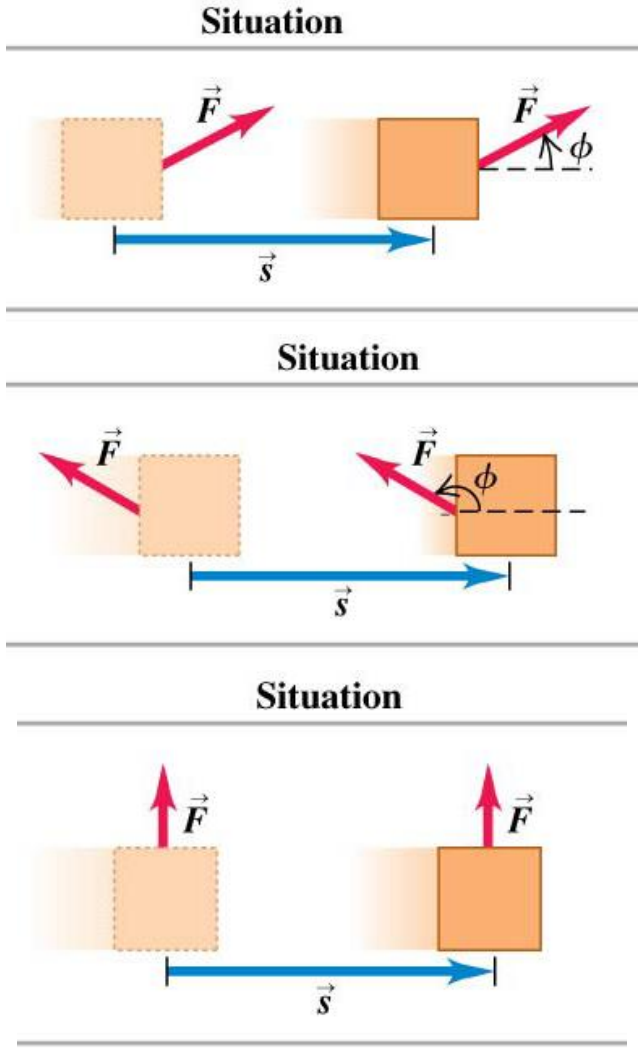
$$= 1.8 \times 10^3\text{ J}$$

# Positive, Negative, or Zero Work

- When the force has a component in the same direction as the displacement ($0° < \Phi < 90°$), the work $W$ is positive. positive work
- When the force has a component opposite to the displacement ($90° < \Phi < 180°$), the work $W$ is negative. work.
- When the force is perpendicular to the displacement ($\Phi = 90°$), the work done by the force is zero

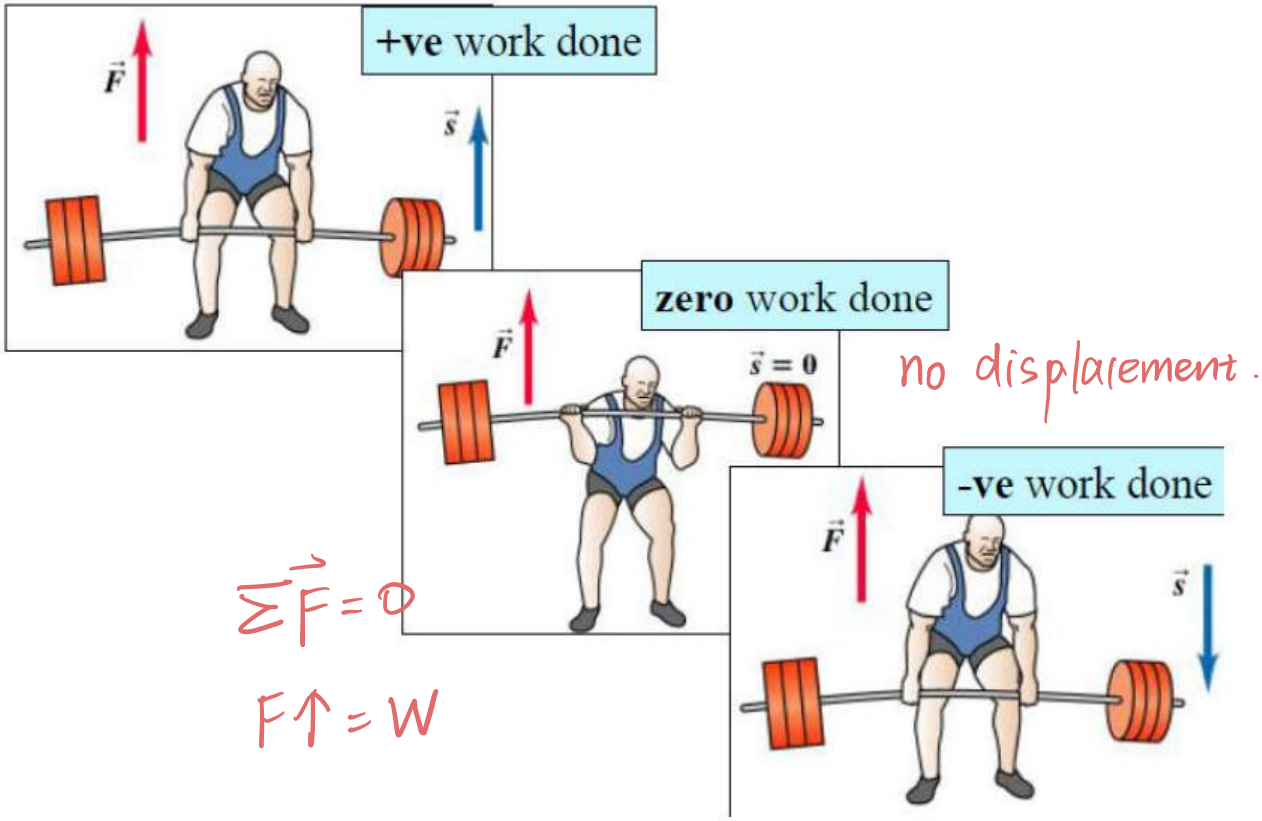
$$W = Fs\cos\phi \quad (6.2)$$

力没有做功.



# Positive, Negative, or Zero Work

**Example:** Work Done by Weight-lifter




no displacement.

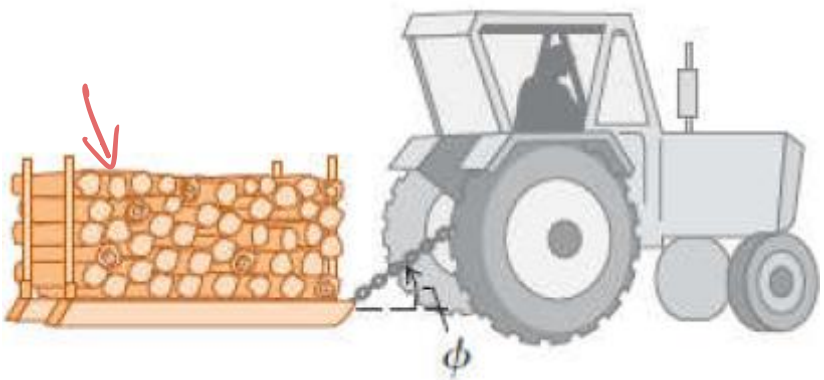
$\Sigma \vec{F} = 0$

$F\uparrow = W$

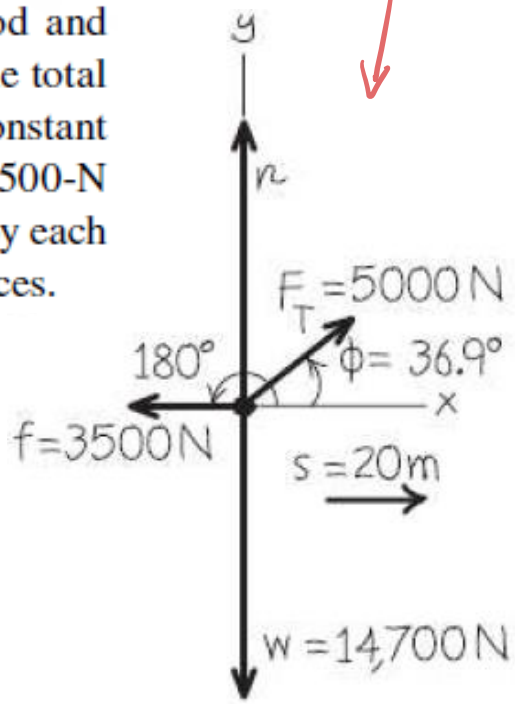
# Positive, Negative, or Zero Work

## EXAMPLE 6.2 WORK DONE BY SEVERAL FORCES

A farmer hitches her tractor to a sled loaded with firewood and pulls it a distance of 20 m along level ground (**Fig. 6.7a**). The total weight of sled and load is 14,700 N. The tractor exerts a constant 5000-N force at an angle of 36.9° above the horizontal. A 3500-N friction force opposes the sled's motion. Find the work done by each force acting on the sled and the total work done by all the forces.




Free body diagram.



# Positive, Negative, or Zero Work

**EXECUTE:** (1) The work $W_w$ done by the weight is zero because its direction is perpendicular to the displacement (compare Fig. 6.4c). For the same reason, the work $W_n$ done by the normal force is also zero. (Note that we don't need to calculate the magnitude $n$ to conclude this.) So $W_w = W_n = 0$.

That leaves the work $W_T$ done by the force $F_T$ exerted by the tractor and the work $W_f$ done by the friction force $f$. From Eq. (6.2),

$$W_T = F_T s\cos\phi = (5000\ \text{N})(20\ \text{m})(0.800) = 80{,}000\ \text{N}\cdot\text{m}$$
$$= 80\ \text{kJ}$$

The friction force $\vec{f}$ is opposite to the displacement, so for this force $\phi = 180°$ and $\cos\phi = -1$. Again from Eq. (6.2),

$$W_f = fs\cos 180° = (3500\ \text{N})(20\ \text{m})(-1) = -70{,}000\ \text{N}\cdot\text{m}$$
$$= -70\ \text{kJ}$$

The total work $W_{tot}$ done on the sled by all forces is the *algebraic* sum of the work done by the individual forces:

$$W_{tot} = W_w + W_n + W_T + W_f = 0 + 0 + 80\ \text{kJ} + (-70\ \text{kJ})$$
$$= 10\ \text{kJ}$$

# Positive, Negative, or Zero Work

(2) In the second approach, we first find the *vector* sum of all the forces (the net force) and then use it to compute the total work. The vector sum is best found by using components. From Fig. 6.7b,

$$\sum F_x = F_T \cos\phi + (-f) = (5000\ \text{N})\cos 36.9° - 3500\ \text{N}$$
$$= 500\ \text{N}$$

$$\sum F_y = F_T \sin\phi + n + (-w)$$
$$= (5000\ \text{N})\sin 36.9° + n - 14{,}700\ \text{N}$$

We don't need the second equation; we know that the $y$-component of force is perpendicular to the displacement, so it does no work. Besides, there is no $y$-component of acceleration, so $\sum F_y$ must be zero anyway. The total work is therefore the work done by the total $x$-component:

$$W_{\text{tot}} = \left(\sum \vec{F}\right) \cdot \vec{s} = \left(\sum F_x\right) s = (500\ \text{N})(20\ \text{m}) = 10{,}000\ \text{J}$$
$$= 10\ \text{kJ}$$

# Energy

能量守恒

- Energy is present in the Universe in various **forms**, e.g. mechanical energy, electromagnetic energy, chemical energy, thermal energy, and nuclear energy.
- Energy can be **transformed** from one form to another.
- **Conservation of energy** is when energy is changed from one form to another, the total amount of energy in the system remains the same, e.g.

  car petrol: chemical -> mechanical + heat

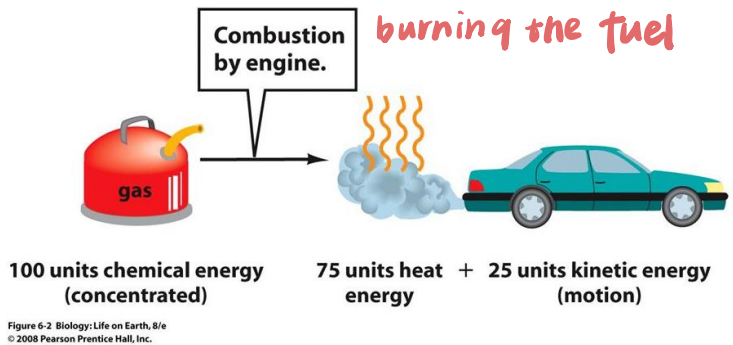
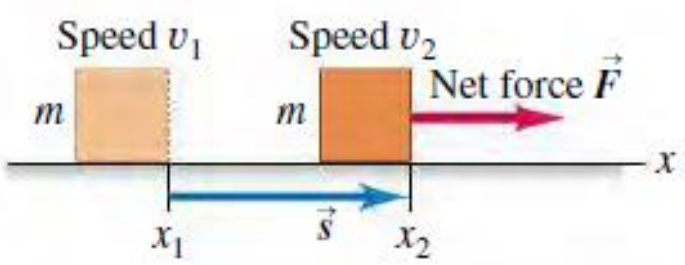



Figure 6-2 Biology: Life on Earth, 8/e
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- The transformation of energy from on form to another is essential part of the study of physics, engineering, chemistry, biology, geology, and astronomy.

# Kinetic Energy

- A box with mass m moves along the positive direction of x-axis under the action of a constant net force and acceleration of magnitude $F$ and $a_x$, respectively.
- Using constant-acceleration equation Eq (2.13), we have:




$$v_2^2 = v_1^2 + 2a_x s$$

$$a_x = \frac{v_2^2 - v_1^2}{2s}$$

- Multiply this equation by $m$ and equate $ma_x$ to the net force $F$, we find:

$$F = ma_x = m\frac{v_2^2 - v_1^2}{2s} \quad \text{and} \quad W = Fs = \tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2 \qquad (6.4)$$

# Kinetic Energy 动能

- The quantity $\frac{1}{2}mv^2$ is called the kinetic energy $K$ of the particle:

Kinetic energy of a particle: $K = \frac{1}{2}mv^2$ (Mass of particle; Speed of particle) (6.5)

- Eq (6.4) can then be rewritten as:

Total work done on particle = work done by net force: $W_{\text{tot}} = K_2 - K_1 = \Delta K$ (Change in kinetic energy) (6.6)

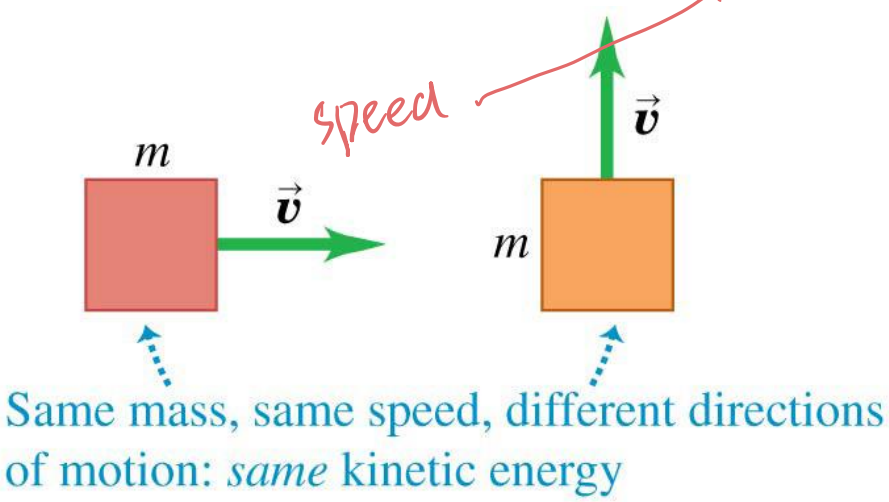
Final kinetic energy: $K_2$; Initial kinetic energy: $K_1$

- Kinetic energy is the energy associated with the motion of a body.
- Kinetic energy is a scalar, and has the same units as work (J).

# Work-Energy Theorem

- Work done by the net force on a particle equals the change in the particle's kinetic energy.
- When $W_{tot}$ is positive, the kinetic energy increases ($K_2 > K_1$) and the particle is going faster at the end $v_2$ of the displacement than at the beginning $v_1$, and vice versa.
- When $W_{tot}$ is zero, kinetic energy stays the same:
- The work–energy theorem by itself tells us only about changes in **speed**, not velocity, since the kinetic energy doesn't depend on the direction of motion.

$$W_{tot} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$



# Work and Kinetic Energy

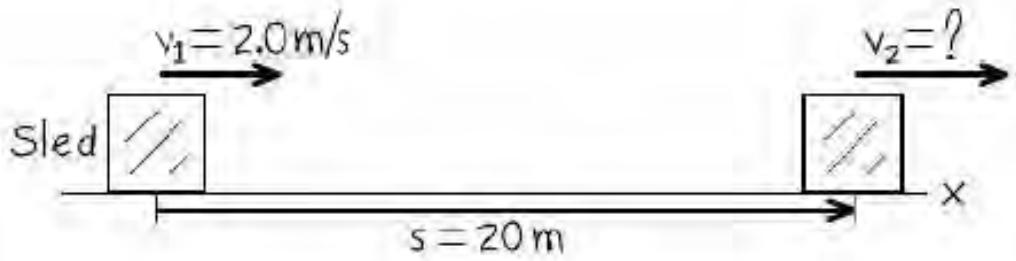
**Example 6.3** **Using work and energy to calculate speed**

Let's look again at the sled in Fig. 6.7 and our results from Example 6.2. Suppose the sled's initial speed $v_1$ is 2.0 m/s. What is the speed of the sled after it moves 20 m?

10kJ

**EXECUTE:** To write expressions for the initial and final kinetic energies, we need the mass of the sled and load. The combined *weight* is 14,700 N, so the mass is

$$m = \frac{w}{g} = \frac{14{,}700\ \text{N}}{9.8\ \text{m/s}^2} = 1500\ \text{kg}$$



# Work and Kinetic Energy

Then the initial kinetic energy $K_1$ is

$$K_1 = \tfrac{1}{2}mv_1^2 = \tfrac{1}{2}(1500\ \text{kg})(2.0\ \text{m/s})^2 = 3000\ \text{kg}\cdot\text{m}^2/\text{s}^2$$

$$= 3000\ \text{J}$$

The final kinetic energy $K_2$ is

$$K_2 = \tfrac{1}{2}mv_2^2 = \tfrac{1}{2}(1500\ \text{kg})v_2^2$$

The work–energy theorem, Eq. (6.6), gives

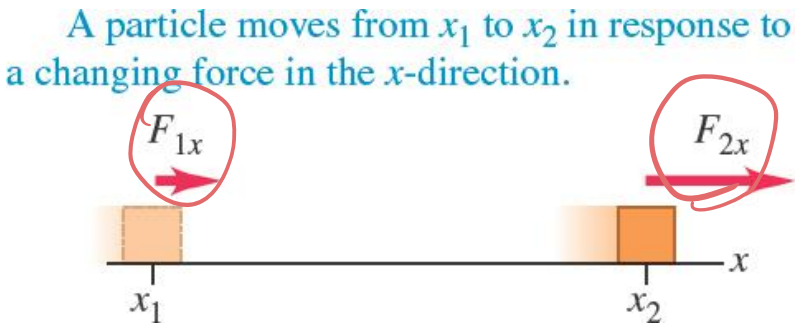
$$K_2 = K_1 + W_{\text{tot}} = 3000\ \text{J} + 10{,}000\ \text{J} = 13{,}000\ \text{J}$$

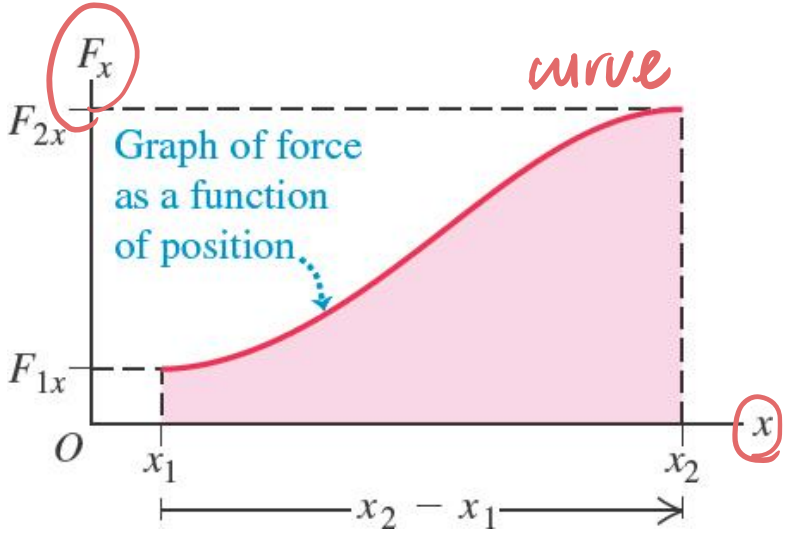
Setting these two expressions for $K_2$ equal, substituting $1\ \text{J} = 1\ \text{kg}\cdot\text{m}^2/\text{s}^2$, and solving for the final speed $v_2$, we find

$$v_2 = 4.2\ \text{m/s}$$

# Work Done by a Varying Force

- Many forces are not constant.
- Suppose a particle moves along the x-axis from $x_1$ to $x_2$.




The force $F_x$ varies with position x.

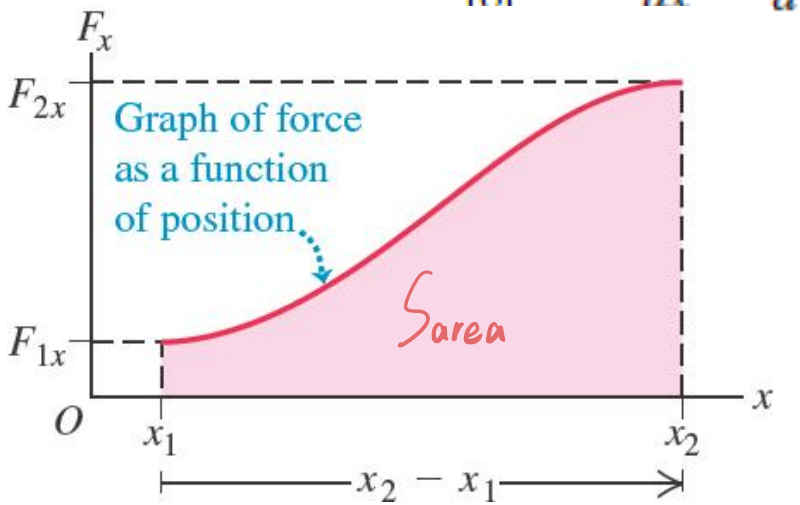
# Work Done by a Varying Force

- For a varying force **F**($x$), consider the work done over a small displacement $\Delta x_a$

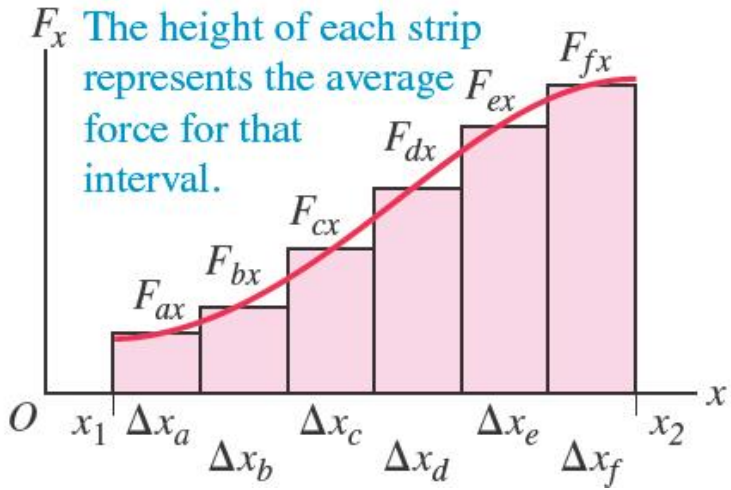
$$W = F_{ax}\Delta x_a$$

- Total work done from the displacement from $x_1$ to $x_2$ is

$$W_{\text{tot}} = F_{ax}\Delta x_a + F_{bx}\Delta x_b + \cdots$$




The force $F_x$ varies with position x.




Over a short displacement $\Delta x$, the force is essentially constant.

# Work Done by a Varying Force

- The work done by the force in the total displacement from $x_1$ to $x_2$ is the integral of $F_x$ from $x_1$ to $x_2$:

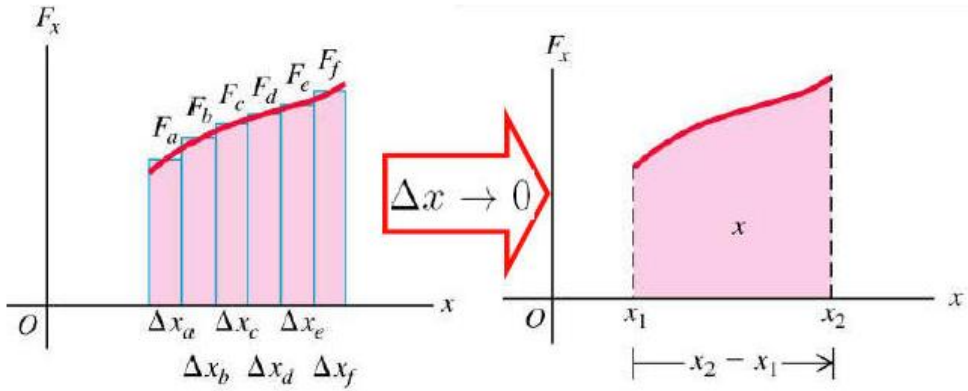
**Work** done on a particle by a **varying $x$-component of force** $F_x$ during **straight-line displacement** along $x$-axis

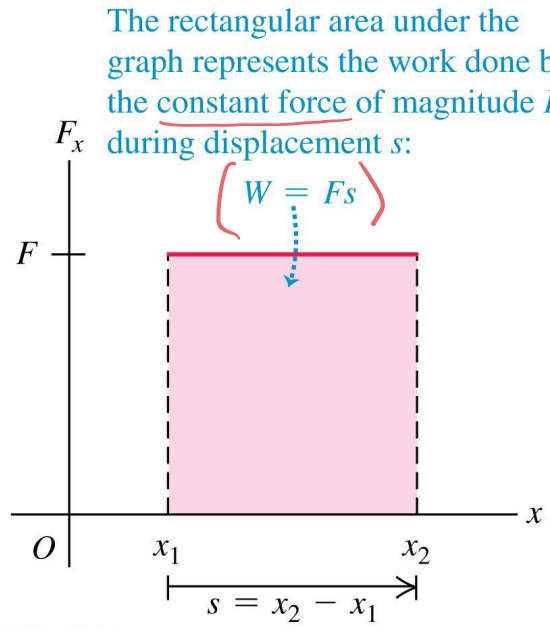
$$W = \int_{x_1}^{x_2} F_x\,dx \qquad (6.7)$$

Upper limit = final position

Integral of $x$-component of force

Lower limit = initial position



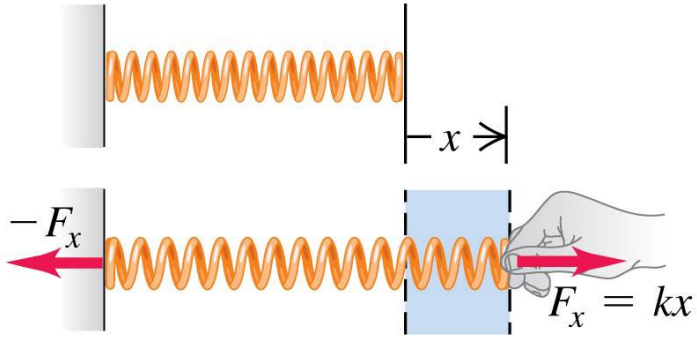




- The work done by the component $F_x$ of the varying force as the particle moves from $x_1$ to $x_2$ is equal to the **area under the curve**.

# Work Done on a Spring 弹簧问题

- Force required to stretch a spring

(Hooke's Law) $F_x = kx$ (6.8)

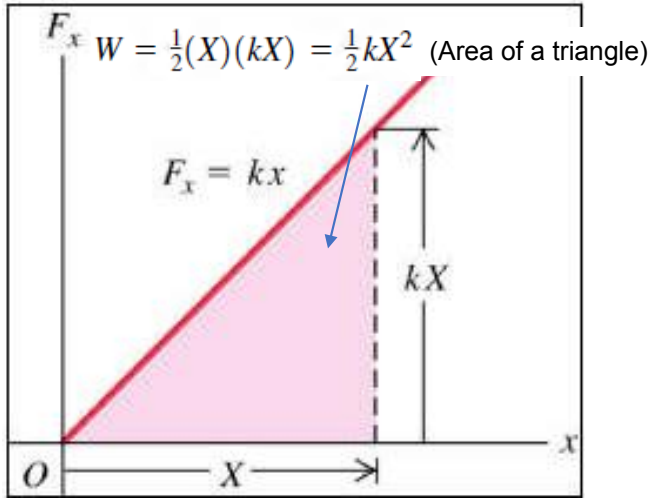



- Work done in stretching spring from $x = 0$ to $X$

$$W = \int_0^X F_x\,dx = \int_0^X kx\,dx = \tfrac{1}{2}kX^2 \qquad (6.9)$$

- Work done in stretching spring from $x = x_1$ to $x = x_2$

$$W = \int_{x_1}^{x_2} F_x\,dx = \int_{x_1}^{x_2} kx\,dx = \tfrac{1}{2}kx_2^2 - \tfrac{1}{2}kx_1^2 \qquad (6.10)$$



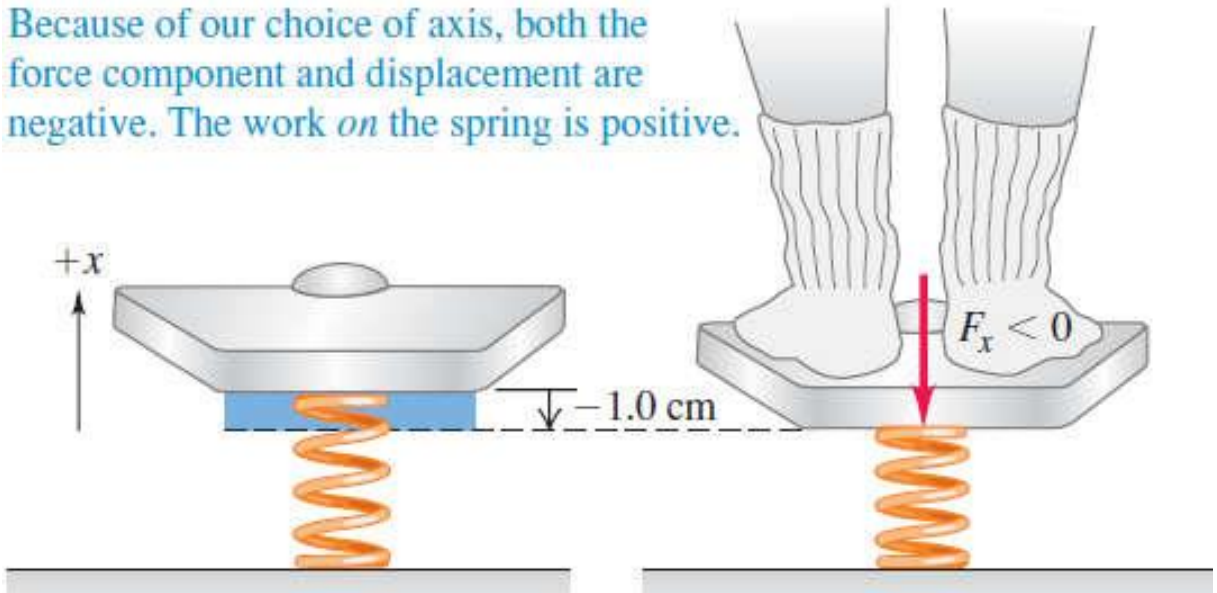

$W = \frac{1}{2}kX^2$ X为与原点之间的距离

# Work Done on a Spring

**Example 6.6** **Work done on a spring scale**

A woman weighing 600 N steps on a bathroom scale that contains a stiff spring (Fig. 6.21). In equilibrium, the spring is compressed 1.0 cm under her weight. Find the force constant of the spring and the total work done on it during the compression.

Because of our choice of axis, both the force component and displacement are negative. The work *on* the spring is positive.



# Work Done on a Spring

**EXECUTE:** The top of the spring is displaced by $x = -1.0\ \text{cm} = -0.010\ \text{m}$, and the woman exerts a force $F_x = -600\ \text{N}$ on the spring. From Eq. (6.8) the force constant is then

$$k = \frac{F_x}{x} = \frac{-600\ \text{N}}{-0.010\ \text{m}} = 6.0 \times 10^4\ \text{N/m}$$

Then, using $x_1 = 0$ and $x_2 = -0.010\ \text{m}$ in Eq. (6.10), we have

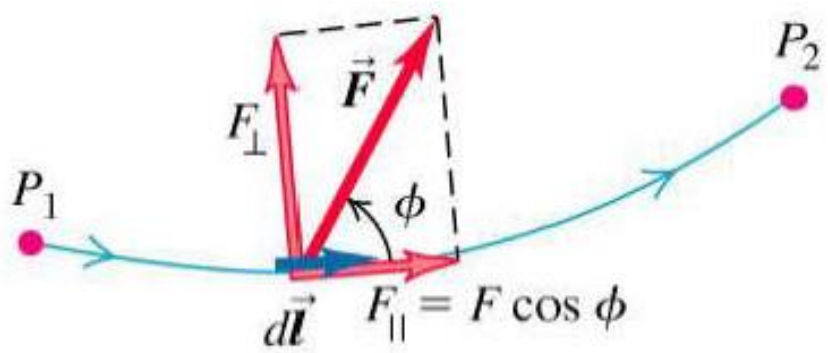
$$W = \tfrac{1}{2}kx_2^2 - \tfrac{1}{2}kx_1^2$$

$$= \tfrac{1}{2}(6.0 \times 10^4\ \text{N/m})(-0.010\ \text{m})^2 - 0 = 3.0\ \text{J}$$

# Work–Energy Theorem for Motion Along a Curve

- In general, for varying force along a curved path, small element of work done *dW* during displacement *dl* is:

$$dW = \vec{F} \cdot d\vec{l} = F\cos\phi\, dl = F_{\parallel}\, dl$$




Only the component of $\vec{F}$ parallel to the displacement, $F_{\parallel} = F\cos\phi$, contributes to the work done by $\vec{F}$.

- Total work done along path P1 to P2 is:

Upper limit = final position

Scalar product (dot product) of $\vec{F}$ and displacement $d\vec{l}$

Work done on a particle by a **varying force** $\vec{F}$ along a **curved path**

$$W = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{l} = \int_{P_1}^{P_2} F\cos\phi\, dl = \int_{P_1}^{P_2} F_{\parallel}\, dl$$

Lower limit = initial position

Angle between $\vec{F}$ and $d\vec{l}$

Component of $\vec{F}$ parallel to $d\vec{l}$

(6.14)

# Power

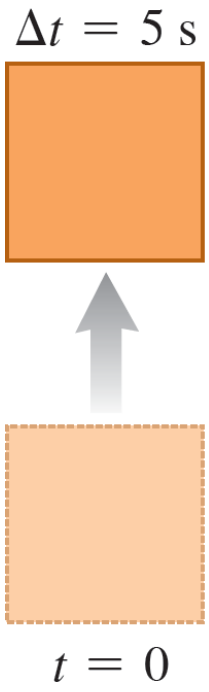
- Power is the time rate at which work is done. Like work and energy, power is a scalar quantity.
  - The average power is: $P_{av} = \frac{\Delta W}{\Delta t}$ ($\Delta W$: Work done during time interval; $\Delta t$: Duration of time interval) (6.15)
  - The instantaneous power is: $P = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$ (Time rate of doing work) (6.16)

$$dW = \vec{F} \cdot d\vec{s} \rightarrow P = \frac{dW}{dt} = \vec{F} \cdot \frac{d\vec{s}}{dt} = \vec{F} \cdot \vec{v} \quad (6.19)$$

- Units for power is Watt, W = 1 J/s
- 1 horsepower (hp) = 746 W
- 1 kWh (kilowatt-hour) = $10^3$ x 3600 Ws = 3.3 MJ (unit for energy)

# Power

- Lifting a Box Slowly



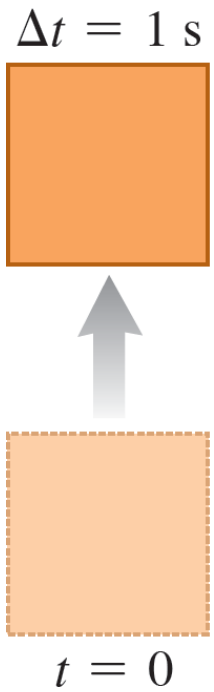

Work you do on the box to lift it in $\Delta t = 5$ s:

$$\Delta W = 100 \text{ J}$$

Your average power output:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t} = \frac{100 \text{ J}}{5 \text{ s}} = 20 \text{ W}$$

- Lifting a Box Quickly




Work you do on the same box to lift it the same distance in $\Delta t = 1$ s:

$$\Delta W = 100 \text{ J}$$

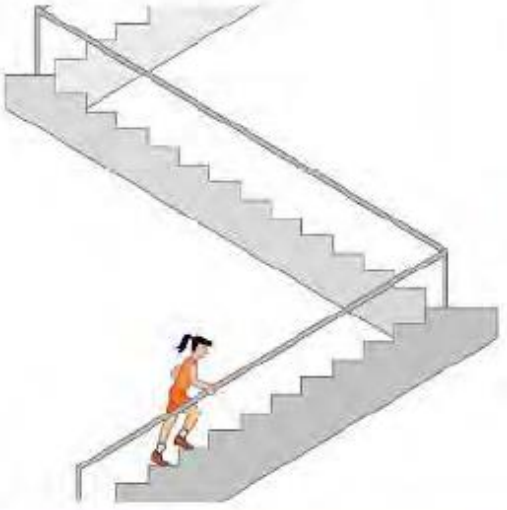
Your average power output:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t} = \frac{100 \text{ J}}{1 \text{ s}} = 100 \text{ W}$$

# Power

## EXAMPLE 6.10 A "POWER CLIMB"

A 50.0-kg marathon runner runs up the stairs to the top of Chicago's 443-m-tall Willis Tower, the second tallest building in the United States (**Fig. 6.28**). To lift herself to the top in 15.0 minutes, what must be her average power output? Express your answer in watts, in kilowatts, and in horsepower.





**Solution:**

Lifting a mass $m$ against gravity requires an amount of work equal to the weight $mg$ multiplied by the height $h$ it is lifted. Hence the work the runner must do is

$$W = mgh = (50.0\ \text{kg})(9.80\ \text{m/s}^2)(443\ \text{m})$$
$$= 2.17 \times 10^5\ \text{J}$$

She does this work in a time 15.0 min = 900 s, so from Eq. (6.15) the average power is

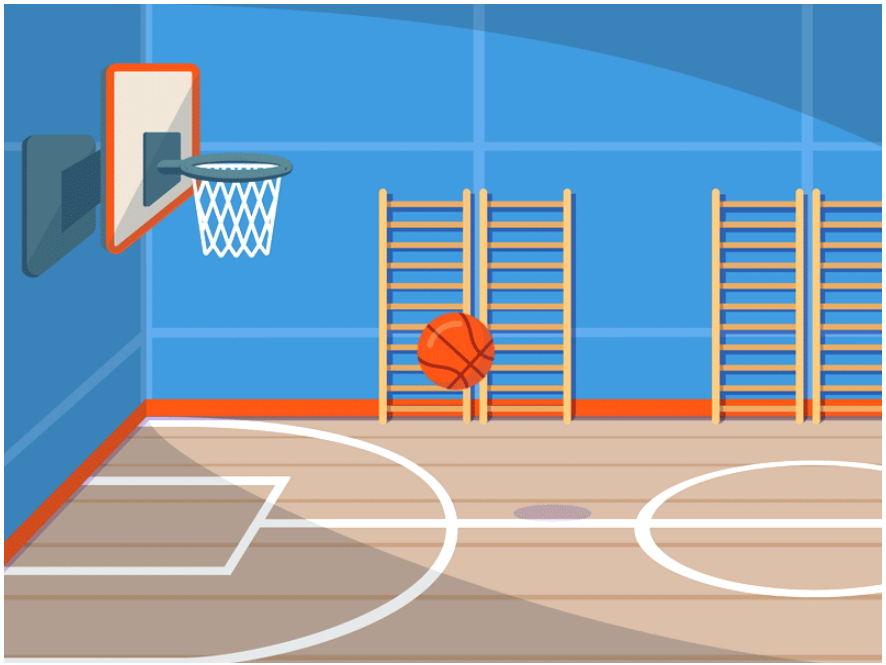
$$P_{\text{av}} = \frac{2.17 \times 10^5\ \text{J}}{900\ \text{s}} = 241\ \text{W} = 0.241\ \text{kW} = 0.323\ \text{hp}$$

# In this next part, you'll learn…

- how to use gravitational potential energy in problems that involve vertical motion.
- how to use elastic potential energy in problems that involve a moving object attached to a stretched or compressed spring.
- the distinction between conservative and nonconservative forces. (Conservative forces always have a corresponding potential-energy function.)
- how to use energy diagrams to understand how an object moves under the influence of a conservative force.

# Gravitational Potential Energy

- Potential energy is associated with the position of bodies in a system.
  - The greater the height of a basketball, the greater the associated gravitational potential energy.
  - As the basketball descends, gravitational potential energy is converted to kinetic energy and the basketball's speed increases.



**Gravitational potential energy** associated with a particle

$$U_{\text{grav}} = mgy$$

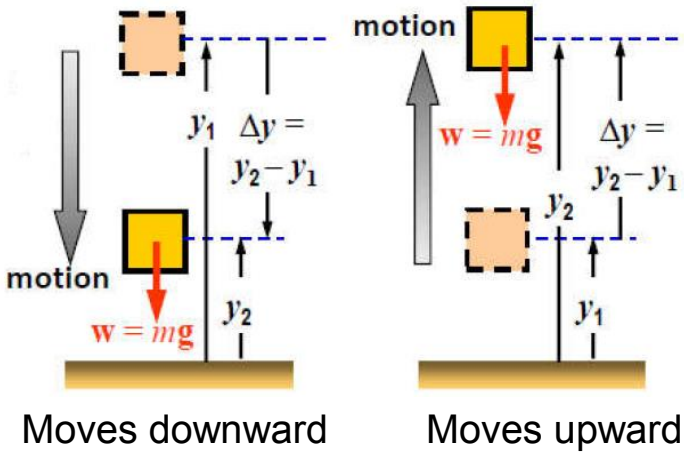
- $m$: Mass of particle
- $g$: Acceleration due to gravity
- $y$: Vertical coordinate of particle ($y$ increases if particle moves upward)

# Gravitational Potential Energy

- Consider an object moving downwards from $y_1$ to $y_2$.
- Work done by gravitational force is given by:

$$W_{\text{grav}} = Fs = w(y_1 - y_2) = mgy_1 - mgy_2 \quad (7.1)$$

- If $m$ is moving upward from $y_1$ to $y_2$, the work done is negative since the weight is opposite to displacement.
- The definition of gravitational potential energy is:



Moves downward Moves upward

Gravitational potential energy associated with a particle

$$U_{\text{grav}} = mgy \quad (7.2)$$

Vertical coordinate of particle (y increases if particle moves upward)

Mass of particle

Acceleration due to gravity

- Combining Eqs (7.1) and (7.2), we get:

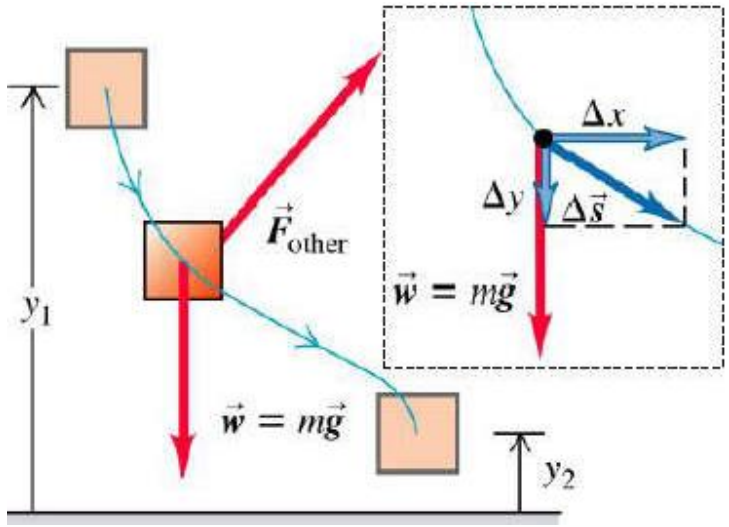
$$W_{\text{grav}} = U_{\text{grav},1} - U_{\text{grav},2} = -(U_{\text{grav},2} - U_{\text{grav},1}) = -\Delta U_{\text{grav}} \quad (7.3)$$

# Gravitational Potential Energy

- If *m* travels along a curved path, we can resolve along the usual x-y direction to find the work done by gravitational force

$$\vec{w} = m\vec{g} = -mg\hat{j}$$

$$\Delta\vec{s} = \Delta x\hat{\imath} + \Delta y\hat{j},$$




$$W_{\text{grav}} = \vec{w}\cdot\Delta\vec{s} = -mg\hat{j}\cdot(\Delta x\hat{\imath} + \Delta y\hat{j}) = -mg\Delta y$$
$$= -mg(y_2 - y_1) = mgy_1 - mgy_2 = U_{\text{grav},1} - U_{\text{grav},2} = -\Delta U_{\text{grav}} \quad (7.3)$$

- Work is unaffected by any horizontal motion, the same expression Eq (7.3) for gravitational potential energy is valid whether the body's path is curved or straight.
- Only the positions of initial and final points in motion are important.

# Conservation of Mechanical Energy

- Mechanical Energy = Kinetic energy + Potential Energy

$$E = K + U$$

机械能守恒

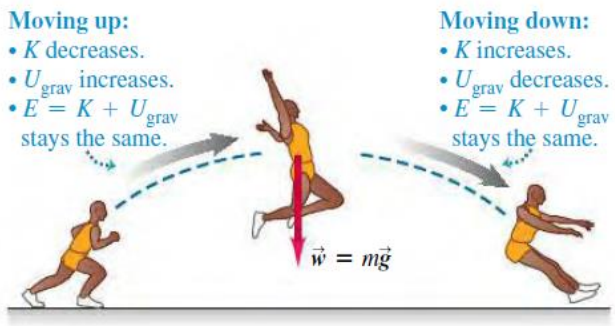
- Conservation of Mechanical Energy
  - ➢ When only the conservative force does work, the total mechanical energy is constant: $E = K + U = Constant$

**If only the gravitational force does work, total mechanical energy is conserved:**

Initial kinetic energy $K_1 = \frac{1}{2}mv_1^2$ — Initial gravitational potential energy $U_{grav,1} = mgy_1$

$$K_1 + U_{grav,1} = K_2 + U_{grav,2} \qquad (7.4)$$

Final kinetic energy $K_2 = \frac{1}{2}mv_2^2$ — Final gravitational potential energy $U_{grav,2} = mgy_2$

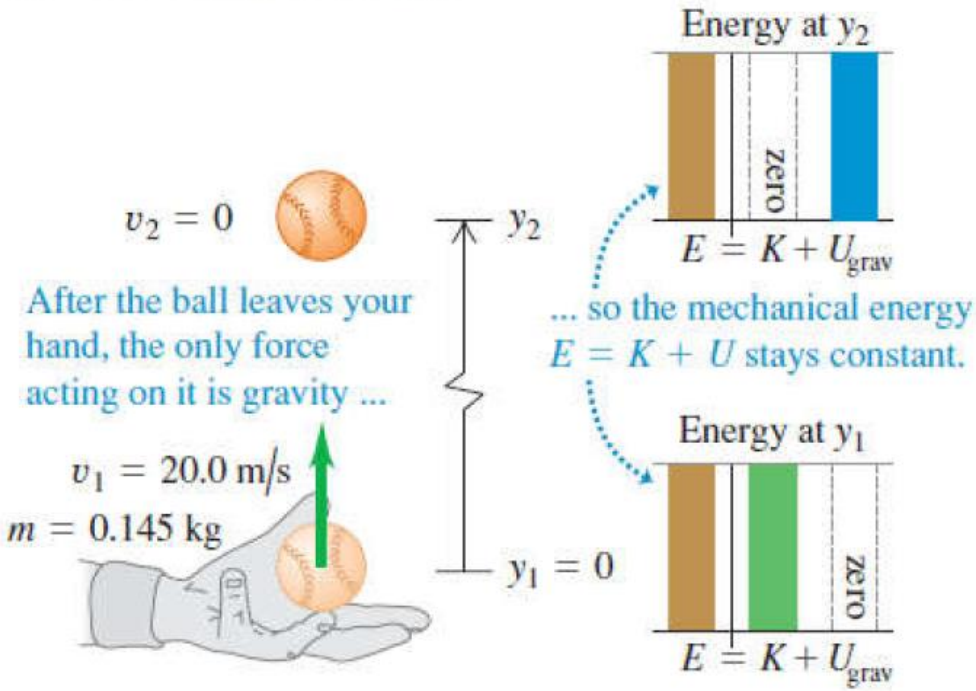



**7.3** While this athlete is in midair, only gravity does work on him (if we neglect the minor effects of air resistance). Mechanical energy $E$—the sum of kinetic and gravitational potential energy—is conserved.

# Conservation of Mechanical Energy

**Example 7.1** You throw a 0.145-kg baseball straight up, giving it an initial velocity of magnitude 20.0 m/s. Find how high it goes, ignoring air resistance.

**7.4** After a baseball leaves your hand, mechanical energy $E = K + U$ is conserved.



# Conservation of Mechanical Energy

**EXECUTE:** We have $y_1 = 0$, $U_{\text{grav},1} = mgy_1 = 0$, and $K_2 = \frac{1}{2}mv_2^2 = 0$. Then Eq. (7.4), $K_1 + U_{\text{grav},1} = K_2 + U_{\text{grav},2}$, becomes

$$K_1 = U_{\text{grav},2}$$

As the energy bar graphs in Fig. 7.4 show, this equation says that the kinetic energy of the ball at point 1 is completely converted to gravitational potential energy at point 2. We substitute $K_1 = \frac{1}{2}mv_1^2$ and $U_{\text{grav},2} = mgy_2$ and solve for $y_2$:

$$\frac{1}{2}mv_1^2 = mgy_2$$

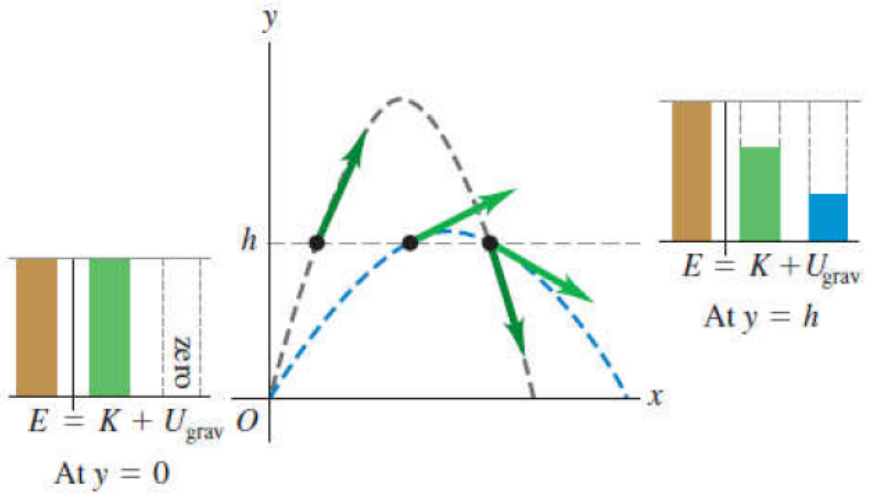
$$y_2 = \frac{v_1^2}{2g} = \frac{(20.0\ \text{m/s})^2}{2(9.80\ \text{m/s}^2)} = 20.4\ \text{m}$$

# Conservation of Mechanical Energy

**Example7.3** A batter hits two identical baseballs with the same initial speed and from the same initial height but at different initial angles. Prove that both balls have the same speed at any height $h$ if air resistance can be neglected.

$$K_1 + U_{\text{grav},1} = K_2 + U_{\text{grav},2}$$

*The work–energy theorem by itself tells us only about changes in **speed**, not velocity, since the kinetic energy doesn't depend on the direction of motion.




**Solution：** The only force acting on each ball after it is hit is its weight. Hence the total mechanical energy for each ball is constant. Figure 7.8 shows the trajectories of two balls batted at the same height with the same initial speed, and thus the same total mechanical energy, but with different initial angles. At all points at the same height the potential energy is the same. Thus the kinetic energy at this height must be the same for both balls, and the speeds are the same.

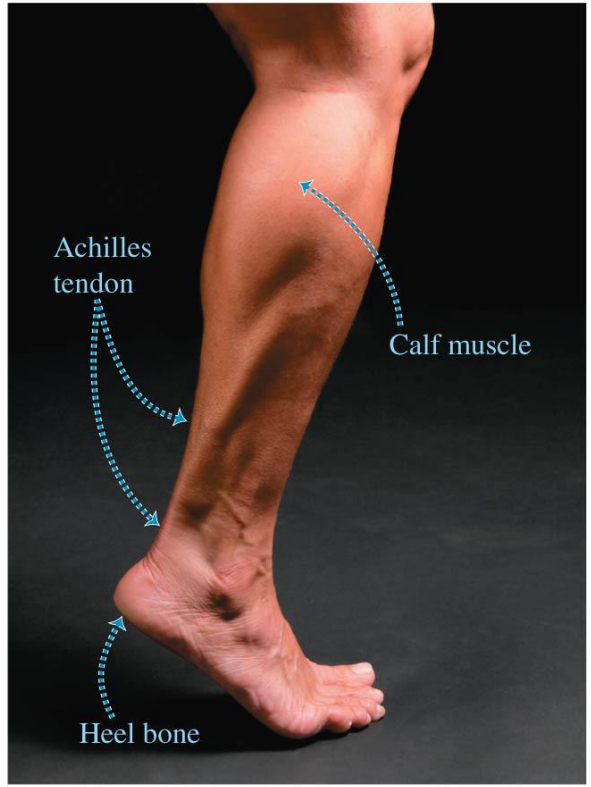
# Elastic Potential Energy 弹性势能

- An object is elastic if it returns to its original shape after being deformed.
- Elastic potential energy is the energy stored in an elastic object, such as a spring:

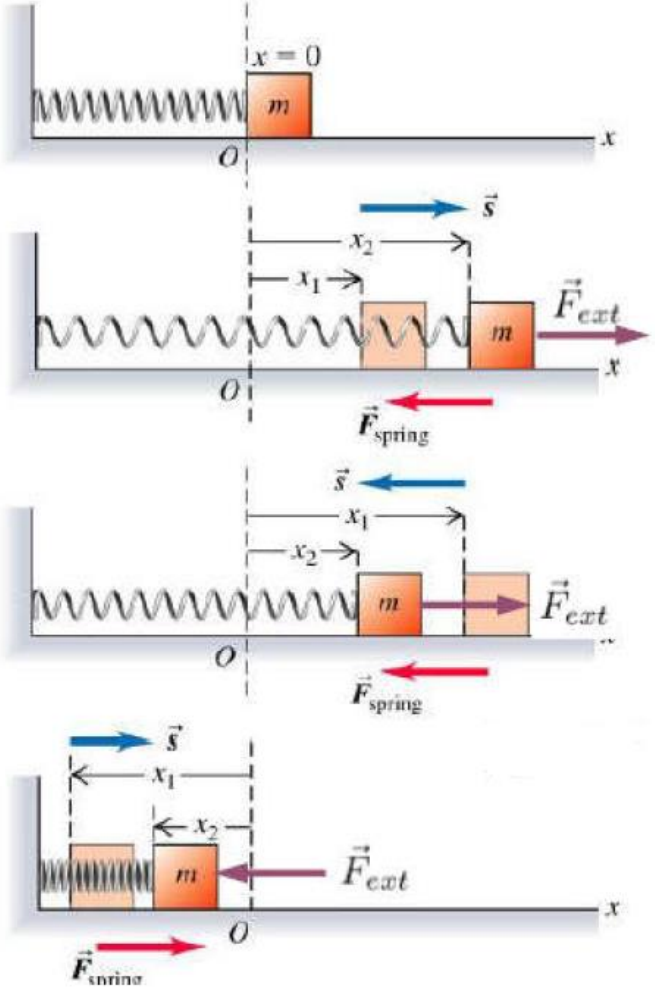
**Elastic potential energy** stored in a spring: $U_{el} = \frac{1}{2}kx^2$

($k$: Force constant of spring; $x$: Elongation of spring ($x > 0$ if stretched, $x < 0$ if compressed))

- The Achilles tendon acts like a natural spring.
- When it stretches and then relaxes, this tendon stores and then releases elastic potential energy.
- This spring action reduces the amount of work your leg muscles must do as you run.



# Elastic Potential Energy




- Recall the work done ***on*** a spring when the block is displaced from $x_1$ to $x_2$:

$$W = \tfrac{1}{2}kx_2^2 - \tfrac{1}{2}kx_1^2 \qquad \text{(work done } on \text{ a spring)} \quad (6.10)$$

- If we let the spring relax while holding one end, we do negative work on it. The ***elastic*** work done ***by*** the spring is just the negative of the work done ***on*** the spring.

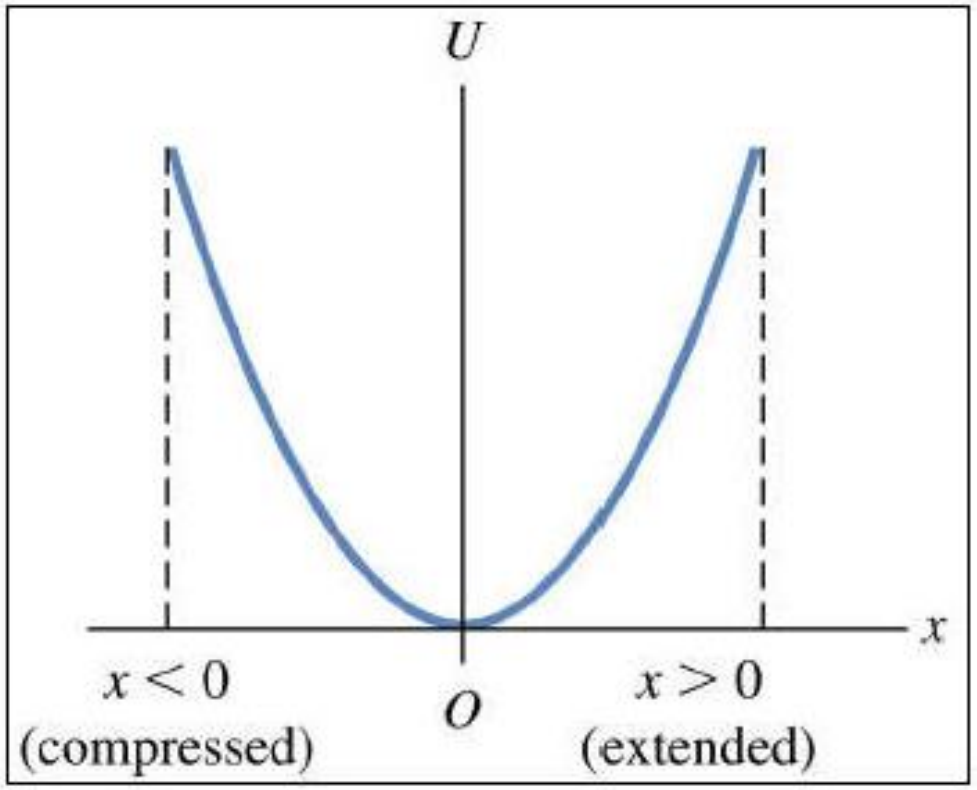
$$W_{el} = \tfrac{1}{2}kx_1^2 - \tfrac{1}{2}kx_2^2 \qquad \text{(work done } by \text{ a spring)} \quad (7.9)$$

$$U_{el} = \tfrac{1}{2}kx^2 \quad (7.10)$$

$$W_{el} = U_{el,1} - U_{el,2} = -\Delta U_{el} \quad (7.11)$$

# Elastic Potential Energy

- Defined as $U_{el} = \frac{1}{2}kx^2$
- There is no negative value!
- Cannot arbitrarily assign zero value! x = *O* is when spring is neither extended nor compressed.
- Will assume that spring constant is the same during compression and extension (unless stated otherwise).



# Situations with Both Kinetic and Elastic Potential Energy

- Work–energy theorem says that $W_{tot} = K_2 - K_1$, if the elastic force is the only force that does work on the body:

$$W_{tot} = W_{el} = U_{el,1} - U_{el,2}$$

- Therefore, if only the elastic force does work, total mechanical energy is conserved:

Initial kinetic energy $K_1 = \frac{1}{2}mv_1^2$ — Initial elastic potential energy $U_{el,1} = \frac{1}{2}kx_1^2$

$$K_1 + U_{el,1} = K_2 + U_{el,2} \quad (7.12)$$

Final kinetic energy $K_2 = \frac{1}{2}mv_2^2$ — Final elastic potential energy $U_{el,2} = \frac{1}{2}kx_2^2$

where

Total work done on particle = work done by net force: $W_{tot} = K_2 - K_1 = \Delta K$ (Change in kinetic energy)

Final kinetic energy: $K_2$; Initial kinetic energy: $K_1$

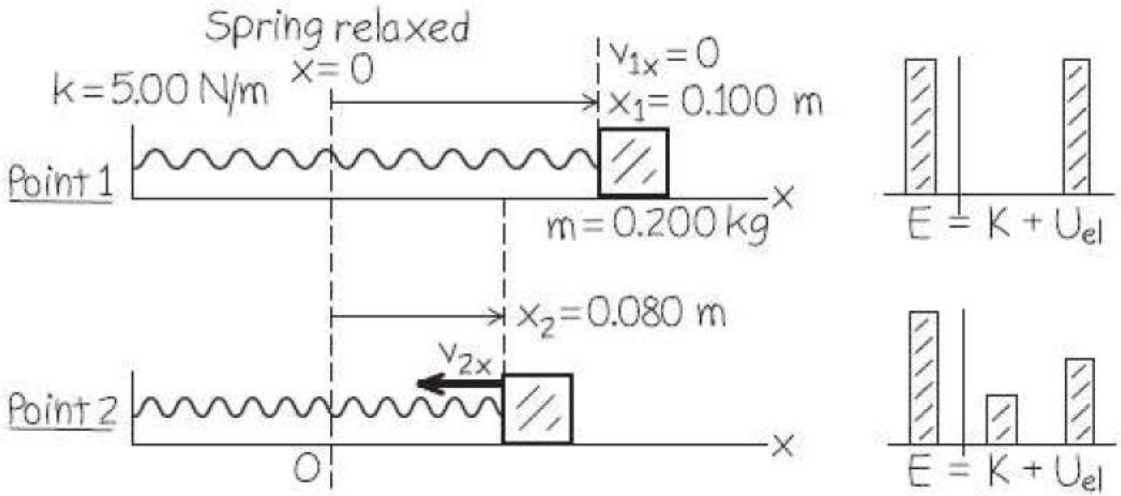
(6.6)

# Example for Elastic Potential Energy

## Example 7.7 Motion with elastic potential energy

A glider with mass $m = 0.200$ kg sits on a frictionless horizontal air track, connected to a spring with force constant $k = 5.00$ N/m. You pull on the glider, stretching the spring 0.100 m, and release it from rest. The glider moves back toward its equilibrium position ($x = 0$). What is its $x$-velocity when $x = 0.080$ m?

**7.16** Our sketches and energy bar graphs for this problem.



# Example for Elastic Potential Energy

**EXECUTE:** The energy quantities are

$$K_1 = \tfrac{1}{2}mv_{1x}^2 = \tfrac{1}{2}(0.200\ \text{kg})(0)^2 = 0$$

$$U_1 = \tfrac{1}{2}kx_1^2 = \tfrac{1}{2}(5.00\ \text{N/m})(0.100\ \text{m})^2 = 0.0250\ \text{J}$$

$$K_2 = \tfrac{1}{2}mv_{2x}^2$$

$$U_2 = \tfrac{1}{2}kx_2^2 = \tfrac{1}{2}(5.00\ \text{N/m})(0.080\ \text{m})^2 = 0.0160\ \text{J}$$

We use Eq. (7.14) with $W_{\text{other}} = 0$ to solve for $K_2$ and then find $v_{2x}$:

$$K_2 = K_1 + U_1 - U_2 = 0 + 0.0250\ \text{J} - 0.0160\ \text{J} = 0.0090\ \text{J}$$

$$v_{2x} = \pm\sqrt{\frac{2K_2}{m}} = \pm\sqrt{\frac{2(0.0090\ \text{J})}{0.200\ \text{kg}}} = \pm 0.30\ \text{m/s}$$

We choose the negative root because the glider is moving in the $-x$-direction. Our answer is $v_{2x} = -0.30$ m/s.

# Situations with Both Gravitational and Elastic Potential Energy

- The total work is the sum of the work done by the gravitational force ($W_{grav}$), the work done by the elastic force ($W_{el}$), and the work done by other forces ($W_{other}$) : $W_{tot} = W_{grav} + W_{el} + W_{other}$
- Then the work–energy theorem gives: $W_{grav} + W_{el} + W_{other} = K_2 - K_1$
- The work done by the gravitational force is $W_{grav} = U_{grav,1} - U_{grav,2}$ and the work done by the spring is $W_{el} = U_{el,1} - U_{el,2}$.
- Hence we can rewrite the work-energy theorem for this most general case as:

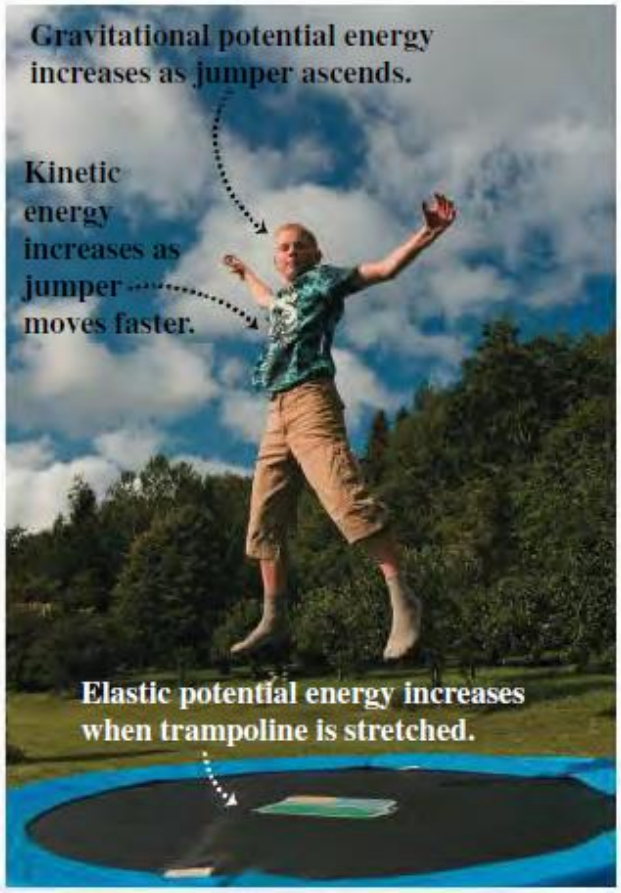
$$K_1 + U_{grav,1} + U_{el,1} + W_{other} = K_2 + U_{grav,2} + U_{el,2} \quad \text{(valid in general)} \quad (7.13)$$

# Situations with Both Gravitational and Elastic Potential Energy

- Or equivalently $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$ (valid in general) [7.14]
- Eq (7.14) is the most general statement of the relationship among kinetic energy, potential energy, and work done by other forces. It says:

  The work done by all forces other than the gravitational force or elastic force equals the change in the real mechanical energy $E = K + U$ of the system, where $U = U_{\text{grav}} + U_{\text{el}} = mgy + \frac{1}{2}kx^2$ is the sum of the gravitational potential energy and the elastic potential energy.
- The "system" is made up of the body of mass *m*, the earth with which it interacts through the gravitational force, and the spring of the force constant *k*.

# Situations with Both Gravitational and Elastic Potential Energy and Work Done by Other Forces




- Trampoline jumping involves an interplay among kinetic energy, gravitational potential energy, and elastic potential energy.
- Due to air resistance and friction forces within the trampoline, mechanical energy is not conserved.
- That's why the bouncing eventually stops unless the jumper does work with his or her legs to compensate for the lost energy.

# Examples Including Work Done by Several Forces

**Example 7.8** **Motion with elastic potential energy and work done by other forces**

Suppose the glider in Example 7.7 is initially at rest at $x = 0$, with the spring unstretched. You then push on the glider with a constant force $\vec{F}$ (magnitude 0.610 N) in the $+x$-direction. What is the glider's velocity when it has moved to $x = 0.100$ m?

**EXECUTE:** The force $\vec{F}$ is constant and in the same direction as the displacement, so the work done by this force is $F\Delta x$. Then the energy quantities are

$$K_1 = 0$$

$$U_1 = \tfrac{1}{2}kx_1^2 = 0$$

$$K_2 = \tfrac{1}{2}mv_{2x}^2$$

$$U_2 = \tfrac{1}{2}kx_2^2 = \tfrac{1}{2}(5.00\ \text{N/m})(0.100\ \text{m})^2 = 0.0250\ \text{J}$$

$$W_{\text{other}} = F\Delta x = (0.610\ \text{N})(0.100\ \text{m}) = 0.0610\ \text{J}$$

# Examples Including Work Done by Several Forces

The initial total mechanical energy is zero; the work done by $\vec{F}$ increases the total mechanical energy to 0.0610 J, of which $U_2 = 0.0250$ J is elastic potential energy. The remainder is kinetic energy. From Eq. (7.14),

$$K_1 + U_1 + W_{\text{other}} = K_2 + U_2$$

$$K_2 = K_1 + U_1 + W_{\text{other}} - U_2$$

$$= 0 + 0 + 0.0610\ \text{J} - 0.0250\ \text{J} = 0.0360\ \text{J}$$

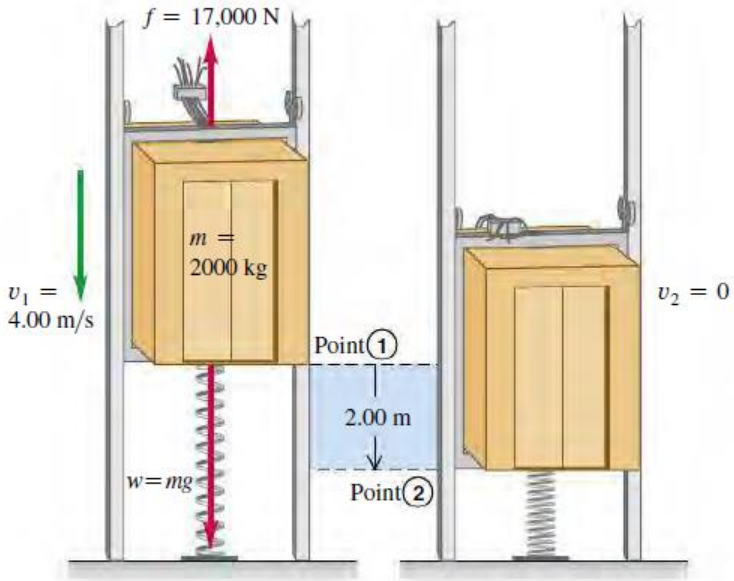
$$v_{2x} = \sqrt{\frac{2K_2}{m}} = \sqrt{\frac{2(0.0360\ \text{J})}{0.200\ \text{kg}}} = 0.60\ \text{m/s}$$

We choose the positive square root because the glider is moving in the $+x$-direction.

# Examples Including Work Done by Several Forces

## Example 7.9 Motion with gravitational, elastic, and friction forces

A 2000-kg (19,600-N) elevator with broken cables in a test rig is falling at 4.00 m/s when it contacts a cushioning spring at the bottom of the shaft. The spring is intended to stop the elevator, compressing 2.00 m as it does so (Fig. 7.17). During the motion a safety clamp applies a constant 17,000-N frictional force to the elevator. What is the necessary force constant $k$ for the spring?



# Examples Including Work Done by Several Forces

**EXECUTE:** The elevator's initial speed is $v_1 = 4.00\ \text{m/s}$, so its initial kinetic energy is

$$K_1 = \tfrac{1}{2}mv_1^2 = \tfrac{1}{2}(2000\ \text{kg})(4.00\ \text{m/s})^2 = 16{,}000\ \text{J}$$

The elevator stops at point 2, so $K_2 = 0$. At point 1 the potential energy $U_1 = U_{\text{grav}} + U_{\text{el}}$ is zero; $U_{\text{grav}}$ is zero because $y_1 = 0$, and $U_{\text{el}} = 0$ because the spring is uncompressed. At point 2 there is both gravitational and elastic potential energy, so

$$U_2 = mgy_2 + \tfrac{1}{2}ky_2^2$$

The gravitational potential energy at point 2 is

$$mgy_2 = (2000\ \text{kg})(9.80\ \text{m/s}^2)(-2.00\ \text{m}) = -39{,}200\ \text{J}$$

The "other" force is the constant 17,000-N friction force. It act opposite to the 2.00-m displacement, so

$$W_{\text{other}} = -(17{,}000\ \text{N})(2.00\ \text{m}) = -34{,}000\ \text{J}$$

## Examples Including Work Done by Several Forces

We put these terms into Eq. (7.14), $K_1 + U_1 + W_{\text{other}} = K_2 + U_2$:

$$K_1 + 0 + W_{\text{other}} = 0 + (mgy_2 + \tfrac{1}{2}ky_2^2)$$

$$k = \frac{2(K_1 + W_{\text{other}} - mgy_2)}{y_2^2}$$

$$= \frac{2[16{,}000\ \text{J} + (-34{,}000\ \text{J}) - (-39{,}200\ \text{J})]}{(-2.00\ \text{m})^2}$$

$$= 1.06 \times 10^4\ \text{N/m}$$

This is about one-tenth the force constant of a spring in an automobile suspension.

# Examples Including Work Done by Several Forces

**EVALUATE:** There might seem to be a paradox here. The elastic potential energy at point 2 is

$$\tfrac{1}{2}ky_2^2 = \tfrac{1}{2}(1.06 \times 10^4\ \text{N/m})(-2.00\ \text{m})^2 = 21{,}200\ \text{J}$$

This is *more* than the total mechanical energy at point 1:

$$E_1 = K_1 + U_1 = 16{,}000\ \text{J} + 0 = 16{,}000\ \text{J}$$

But the friction force *decreased* the mechanical energy of the system by 34,000 J between points 1 and 2. Did energy appear from nowhere? No. At point 2, which is below the origin, there is also *negative* gravitational potential energy $mgy_2 = -39{,}200$ J. The total mechanical energy at point 2 is therefore not 21,200 J but

$$\begin{aligned} E_2 &= K_2 + U_2 = 0 + \tfrac{1}{2}ky_2^2 + mgy_2 \\ &= 0 + 21{,}200\ \text{J} + (-39{,}200\ \text{J}) = -18{,}000\ \text{J} \end{aligned}$$

This is just the initial mechanical energy of 16,000 J minus 34,000 J lost to friction.

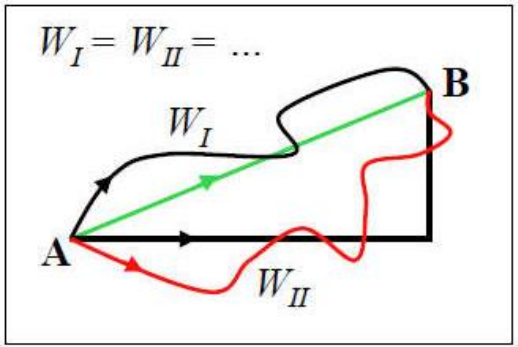
Will the elevator stay at the bottom of the shaft? At point 2 the compressed spring exerts an upward force of magnitude $F_{\text{spring}} = (1.06 \times 10^4\ \text{N/m})(2.00\ \text{m}) = 21{,}200\ \text{N}$, while the downward force of gravity is only $w = mg = (2000\ \text{kg})(9.80\ \text{m/s}^2) = 19{,}600\ \text{N}$. If there were no friction, there would be a net upward force of 21,200 N − 19,600 N = 1600 N, and the elevator would rebound. But the safety clamp can exert a kinetic friction force of 17,000 N, and it can presumably exert a maximum static friction force greater than that. Hence the clamp will keep the elevator from rebounding.

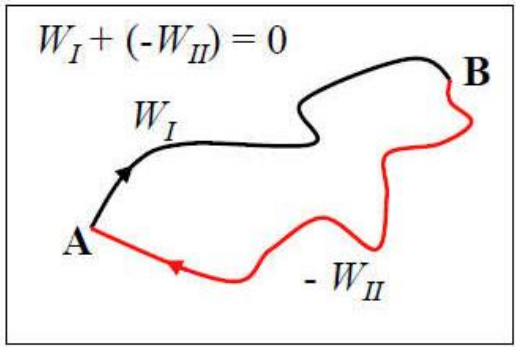
# Conservative Force

- A force that offers the opportunity of two-way conversion between kinetic and potential energies is called a conservative force, i.e. the gravitational force and the spring force.
- A force is conservative if the work it does on a particle moving between any two points is **independent** of the path taken by the particle.

Equivalent statements

- Work done by a conservative force exerted on a particle moving through any **closed path** is zero.



# Conservative Force

1. It can be expressed as the difference between the initial and final values of a potential-energy function.
2. It is reversible. Anything that we deposit in the energy “bank” can later be withdrawn without loss.
3. It is independent of the path of the body and depends on only the starting and ending points.
4. When the starting and ending points are the same, the total work is zero.

# Nonconservative Force

- A force that is not conservative is called a **nonconservative force**.
  - The work done by a nonconservative force **cannot be** represented by a potential-energy function.
- Some nonconservative forces, such as kinetic friction or fluid resistance, cause mechanical energy to be lost or dissipated; a force of this kind is called a **dissipative force**.
- There are also nonconservative forces that increase mechanical energy.
  - For example, the fragments of an exploding firecracker fly off with very large kinetic energy due to a chemical reaction of gunpowder with oxygen. It is nonconservative because the process is not reversible. The fragments never spontaneously reassemble themselves into a complete firecracker!

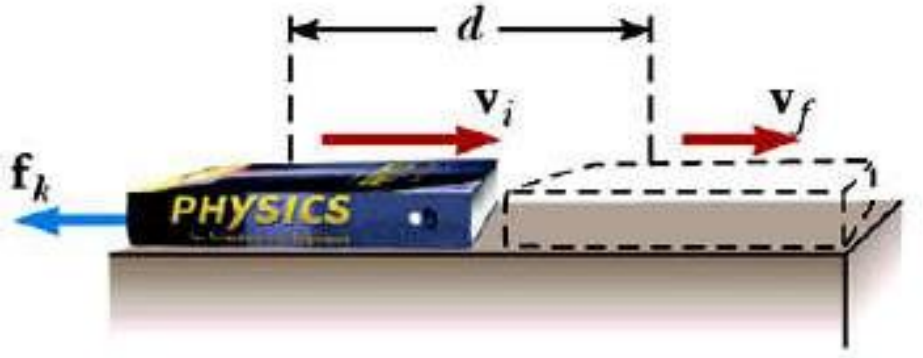
# Nonconservative Force

Nonconservative Force: Friction Force

- Suppose a book is given an initial velocity $v_i$ slides through a rough horizontal table surface for a distance d before reaching a final velocity $v_f$

$$\Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = -f_k d$$

- Loss in kinetic energy is $f_k d$, which is the energy dissipated by the force of kinetic friction. Part of the energy is transferred to the internal energy of the book, and part to the surface.



# Summary

- Work and Kinetic Energy are discussed
  - Work done by constant force
  - Work done by a varying force
  - Work done by/on a spring
  - Kinetic Energy
  - **The Work-Energy Theorem**
  - Power
- Potential Energy are discussed
  - Gravitational Potential Energy
  - Elastic Potential Energy
  - **Conservation of Mechanical Energy**
  - Conservative Force and Non-conservative Force
- Textbook
  - Chapter 6： Work and Kinetic Energy
  - Chapter 7： Potential Energy